

preprocessing

April 11, 2025

Libraries

```
[1]: import pandas as pd
import numpy as np
from matplotlib import pyplot as plt
import seaborn as sns
from statsmodels.stats.outliers_influence import variance_inflation_factor
from statsmodels.tools.tools import add_constant
```

Original Dataset

```
[2]: data = pd.read_stata('/workspaces/malnutrition/NPKR82FL.DTA')
data.head()
```

```
[2]:
```

	caseid	bidx	v000	v001	v002	v003	v004	v005	v006	v007	\	
0	1	1	2	1	NP8	1	1	2	1	916093	magh	2078
1	1	8	2	1	NP8	1	8	2	1	916093	magh	2078
2	1	8	2	2	NP8	1	8	2	1	916093	magh	2078
3	1	9	3	1	NP8	1	9	3	1	916093	magh	2078
4	1	19	1	1	NP8	1	19	1	1	916093	magh	2078

	...	s604c	s607ea	s607eb	s607ec	s607ed	s607ex	s607f	s607g	s615f	s631o
0	...	no	no	no	no	no	no	no	no	NaN	NaN
1	...	no	no	yes	no	no	no	NaN	no	NaN	no
2	...	NaN	no	yes	no	no	no	NaN	no	NaN	no
3	...	no	no	no	no	no	no	yes	no	NaN	no
4	...	no	yes	yes	no	no	no	NaN	no	NaN	NaN

[5 rows x 1422 columns]

```
[3]: data.shape
```

```
[3]: (5372, 1422)
```

Selected Features

```
[4]: df1 = data.copy()
```

```
df1 = df1[['b19', 'v106', 'v525', 'v228', 'v190', 'v102', 'v101', 'v445',
↳ 'b11', 'v137', 'v208', 'v130', 'v131', 'm19', 'b4', 'm4', 'v012', 'v212',
↳ 'v729', 'v714', 'v481', 'hw71', 'hw70', 'hw72']]
df1.rename(columns={'b19': 'Child_age', 'v106': 'Mother_education', 'v525':
↳ 'Age_first_sex', 'v228': 'Pregnancy_terminated', 'v190': 'Wealth_index',
↳ 'v102': 'Place_residence', 'v101': 'Region', 'v445': 'BMI', 'b11':
↳ 'Birth_interval', 'v137': 'Children_under5', 'v208':
↳ 'Total_children_ever_born', 'v130': 'Religion', 'v131': 'Ethnicity', 'm19':
↳ 'Birth_weight', 'b4': 'Child_sex', 'm4': 'Breastfeeding_duration', 'v012':
↳ 'Mother_age_current', 'v212': 'Mother_age_at_first_birth', 'v729':
↳ 'Father_education', 'v714': 'Mother_working', 'v481': 'Health_insurance',
↳ 'hw71': 'Underweight', 'hw70': 'Stunting', 'hw72': 'Wasting'}, inplace=True)
df1.head()
```

```
[4]:      Child_age  Mother_education  Age_first_sex  Pregnancy_terminated  Wealth_index \
0           41             basic           14.0                        no      poorest
1           28             basic           29.0                        no      poorest
2           42             basic           29.0                        no      poorest
3           17             basic           14.0                        no      poorest
4           40        secondary           17.0                       yes      poorer

      Place_residence  Region      BMI  Birth_interval  Children_under5  ... \
0             rural  koshi      NaN           36.0           1  ...
1             rural  koshi      NaN           13.0           2  ...
2             rural  koshi      NaN              NaN           2  ...
3             rural  koshi  2200.0           207.0           1  ...
4             rural  koshi  2510.0           19.0           2  ...

      Child_sex  Breastfeeding_duration  Mother_age_current  \
0      female              NaN              29
1      female  still breastfeeding              33
2      female              NaN              33
3      female  still breastfeeding              34
4       male              NaN              26

      Mother_age_at_first_birth      Father_education  Mother_working  \
0              17      complete primary              yes
1              30      incomplete primary              no
2              30      incomplete primary              no
3              16      incomplete secondary              no
4              21      complete primary              yes

      Health_insurance  Underweight  Stunting  Wasting
0              no              NaN      NaN      NaN
1              no              NaN      NaN      NaN
2              no              NaN      NaN      NaN
3              no          -65.0    -111.0    -19.0
```

```
4          no          -185.0   -311.0   -7.0
```

```
[5 rows x 24 columns]
```

24 features are selected out of 1422 features on the basis of existing literature, which are:

‘Child_age’, ‘Mother_education’, ‘Age_first_sex’, ‘Pregnancy_terminated’, ‘Wealth_index’, ‘Place_residence’, ‘Region’, ‘BMI’, ‘Birth_interval’, ‘Children_under5’, ‘Total_children_ever_born’, ‘Religion’, ‘Ethnicity’, ‘Birth_weight’, ‘Child_sex’, ‘Breastfeeding_duration’, ‘Mother_age_current’, ‘Mother_age_at_first_birth’, ‘Father_education’, ‘Mother_working’, ‘Health_insurance’, ‘Underweight’, ‘Stunting’, ‘Wasting’

```
[5]: df1.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 5372 entries, 0 to 5371
Data columns (total 24 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   Child_age                            5372 non-null   int8
1   Mother_education                    5372 non-null   category
2   Age_first_sex                      5371 non-null   category
3   Pregnancy_terminated                5372 non-null   category
4   Wealth_index                       5372 non-null   category
5   Place_residence                    5372 non-null   category
6   Region                             5372 non-null   category
7   BMI                                2736 non-null   category
8   Birth_interval                     3172 non-null   float64
9   Children_under5                    5372 non-null   int8
10  Total_children_ever_born            5372 non-null   category
11  Religion                           5372 non-null   category
12  Ethnicity                          5372 non-null   category
13  Birth_weight                       3177 non-null   category
14  Child_sex                          5372 non-null   category
15  Breastfeeding_duration              3177 non-null   category
16  Mother_age_current                 5372 non-null   int8
17  Mother_age_at_first_birth          5372 non-null   int8
18  Father_education                   5322 non-null   category
19  Mother_working                     5372 non-null   category
20  Health_insurance                   5372 non-null   category
21  Underweight                        2601 non-null   category
22  Stunting                           2592 non-null   category
23  Wasting                            2592 non-null   category
dtypes: category(19), float64(1), int8(4)
memory usage: 291.5 KB
```

```
[6]: # df1.loc[1369, 'Ethnicity']
```

```
[7]: df2 = df1.copy()
df2.drop(index = 3251, inplace=True) # BMI 9998.0
df2.drop(index = [658,730, 4417], inplace = True) # Religion 96
df2.drop(index = [1369, 602, 603], inplace=True) # Ethnicity 96
df2 = df2[~df2['Birth_weight'].isin(['not weighed at birth', 'don't know'])] #
↳Birth_weight 9998 or 9996
df2 = df2[~df2['Father_education'].isin(['don't know'])]
df2 = df2[~df2['Underweight'].isin(['flagged cases'])]
df2 = df2[~df2['Wasting'].isin(['height out of plausible limits', 'flagged_
↳cases'])] # Wasting 9998 or 9996
df2 = df2[~df2['Stunting'].isin(['flagged cases'])]

df2['Religion'] = df2['Religion'].cat.remove_unused_categories()
df2['Ethnicity'] = df2['Ethnicity'].cat.remove_unused_categories()
```

```
[8]: df2.shape
```

```
[8]: (4636, 24)
```

```
[9]: df2 = df2.dropna(subset=['Underweight', 'Stunting', 'Wasting'])
```

```
[10]: df2.isnull().sum()
```

```
[10]: Child_age          0
Mother_education      0
Age_first_sex        1
Pregnancy_terminated  0
Wealth_index         0
Place_residence      0
Region              0
BMI                 0
Birth_interval       966
Children_under5      0
Total_children_ever_born 0
Religion            0
Ethnicity           0
Birth_weight        1025
Child_sex           0
Breastfeeding_duration 1025
Mother_age_current   0
Mother_age_at_first_birth 0
Father_education     24
Mother_working       0
Health_insurance     0
Underweight         0
Stunting            0
Wasting             0
```

dtype: int64

Numeric Features

```
[11]: numeric_cols = ['Child_age', 'BMI', 'Birth_interval', 'Age_first_sex',  
    ↪ 'Birth_weight', 'Mother_age_current', 'Mother_age_at_first_birth',  
    ↪ 'Children_under5', 'Total_children_ever_born', 'Underweight', 'Stunting',  
    ↪ 'Wasting']  
df2[numeric_cols] = df2[numeric_cols].apply(pd.to_numeric, errors = 'coerce')
```

WHO reference standard

```
[12]: df2['BMI'] = df2['BMI']/100  
df2["Underweight"] = df2["Underweight"]/100  
df2['Stunting'] = df2["Stunting"]/100  
df2['Wasting'] = df2["Wasting"]/100  
  
df2['Underweight'] = [0 if -2 <= val <= 2 else 1 for val in df2['Underweight']]  
    ↪ # Weight-for-Age (WAZ) → Underweight  
df2['Stunting'] = [0 if -2 <= val <= 2 else 1 for val in df2['Stunting']] #  
    ↪ Height-for-Age (HAZ) → Stunting  
df2['Wasting'] = [0 if -2 <= val <= 2 else 1 for val in df2['Wasting']] #  
    ↪ Weight-for-Height (WHZ) → Wasting
```

Z-score Concept (Standard Deviation Score)

Malnutrition classification is based on how far a child's anthropometric measurement deviates from the WHO reference standard for children of the same age and sex. This deviation is quantified using the *Z-score*, also known as the *standard deviation score*.

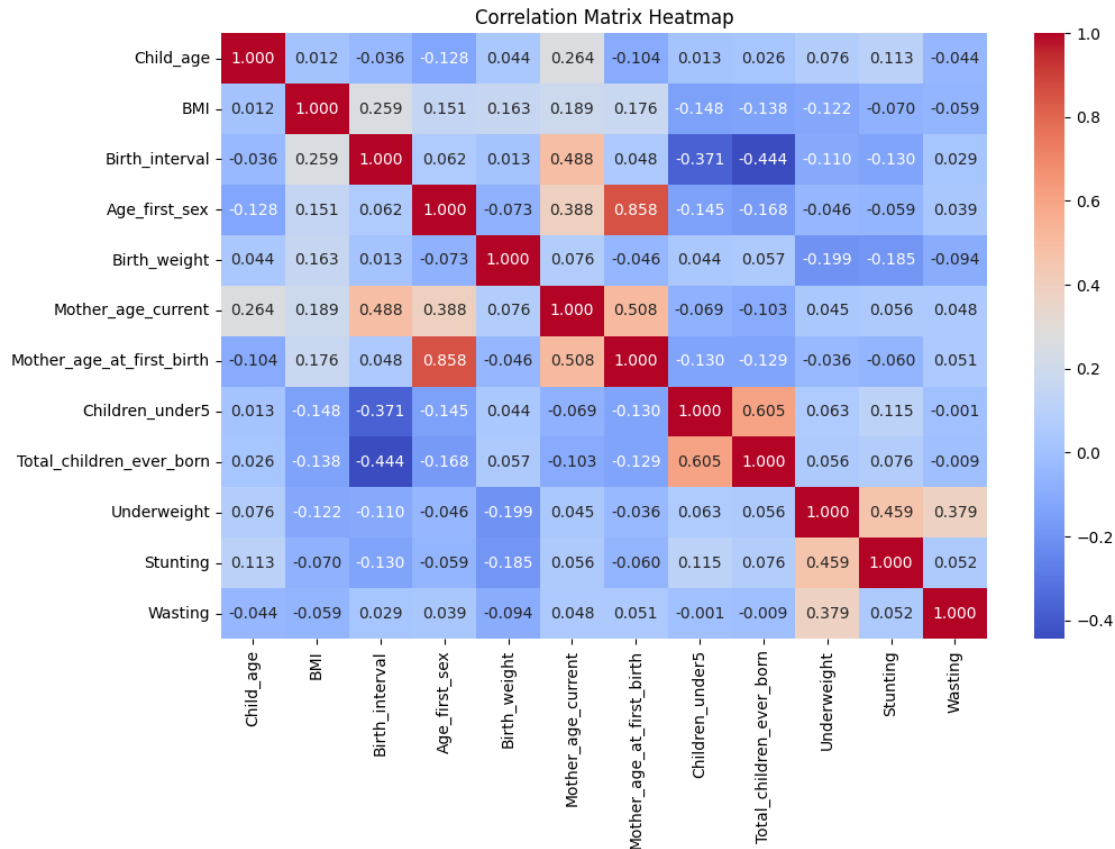
A Z-score tells you how many standard deviations (SDs) a measurement is from the reference population median.

$$Z = \frac{(X - \mu)}{\sigma}$$

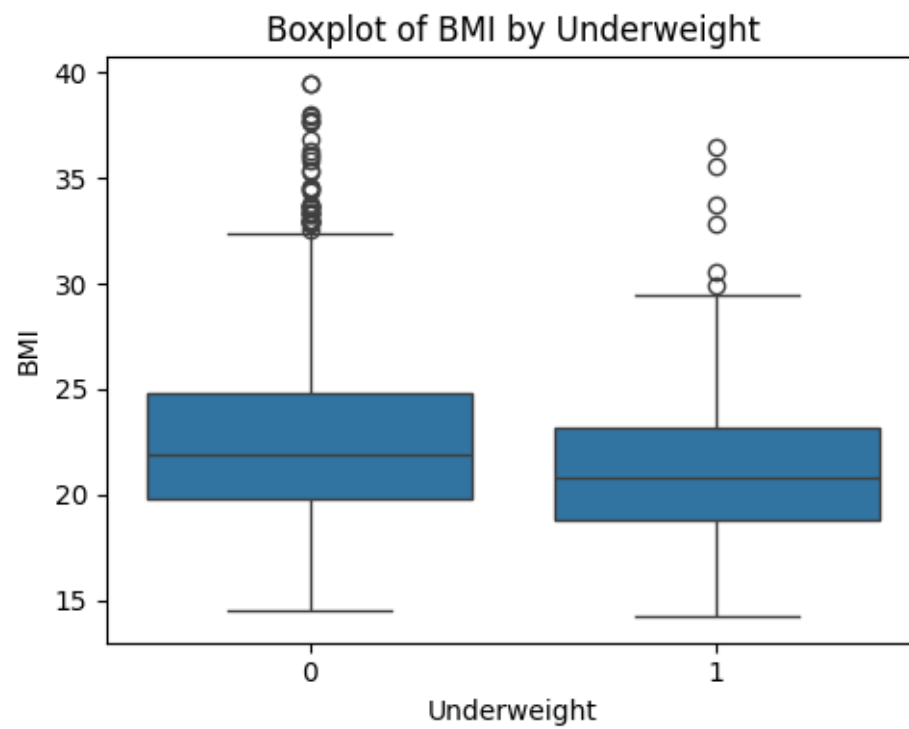
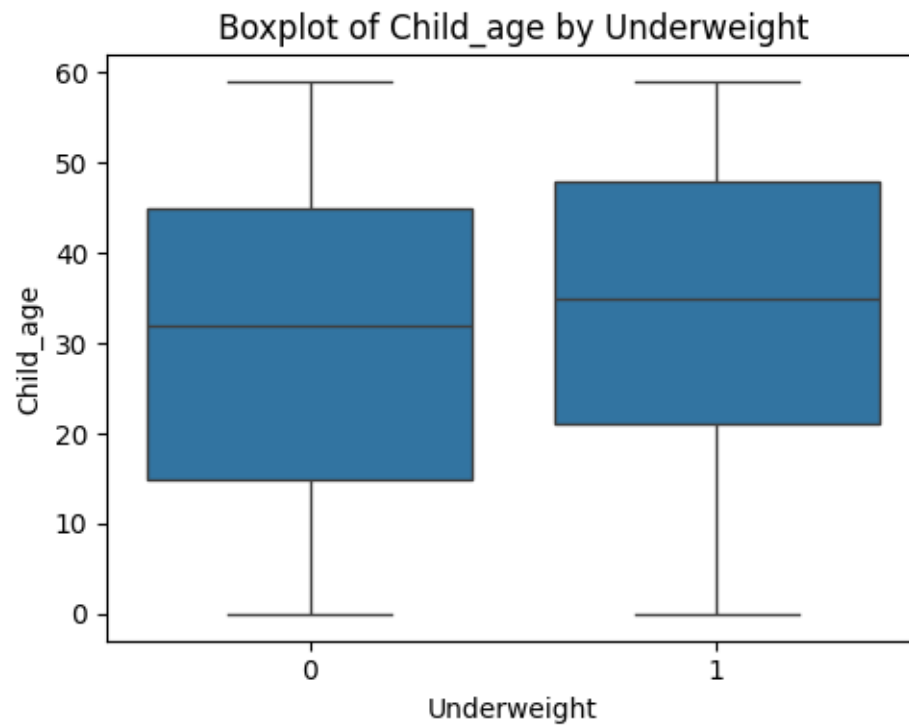
Where, - X = Observed measurement (e.g. child's weight or height) - μ = Median of the reference population (matched by age and sex) - σ = Standard deviation of the reference population

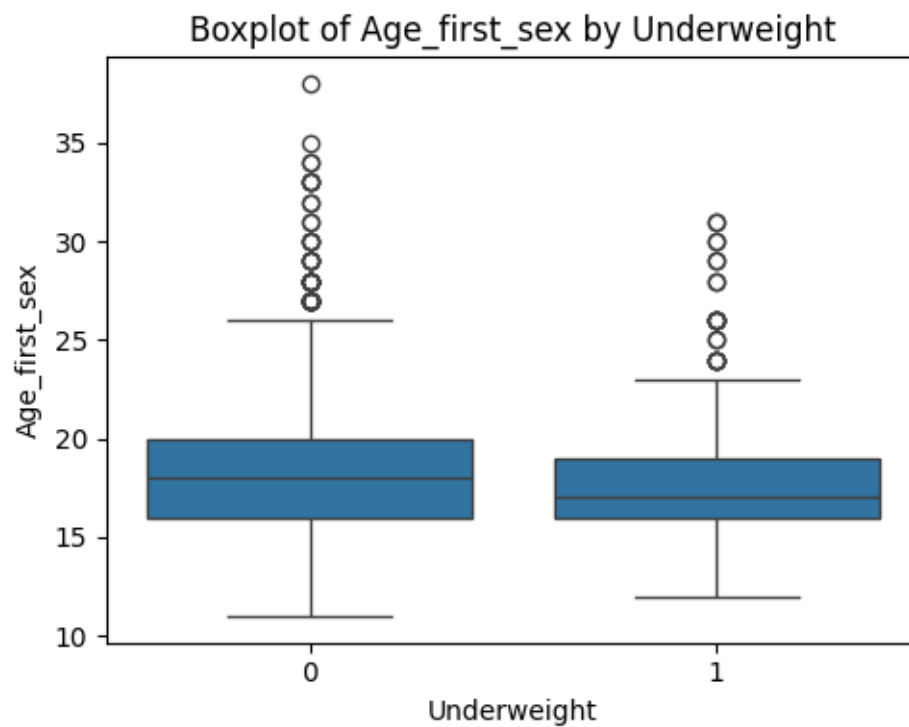
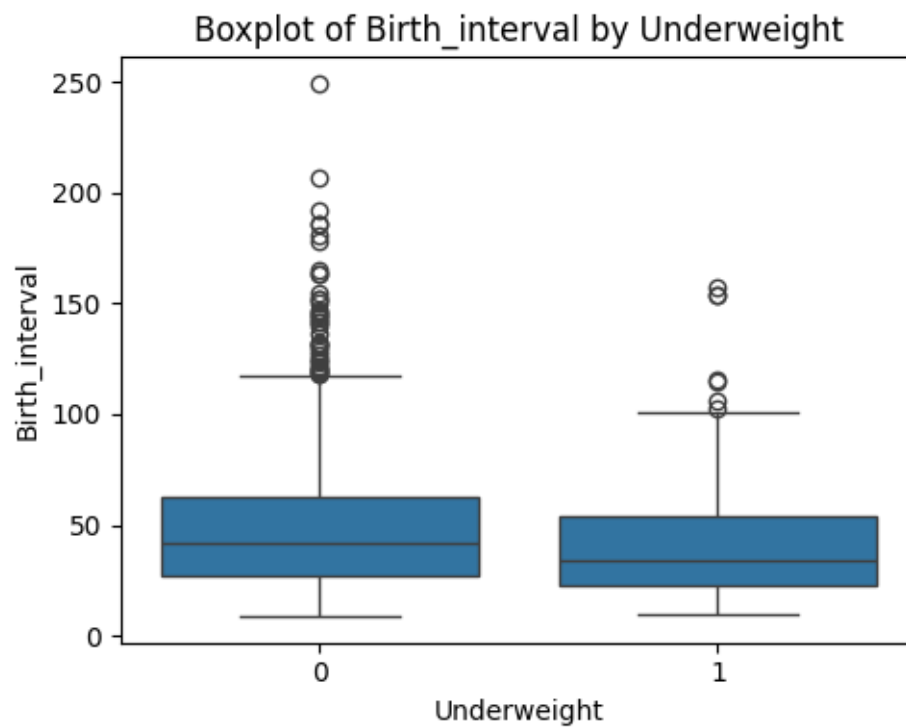
0.1 Feature Importance Check for Numerical Features

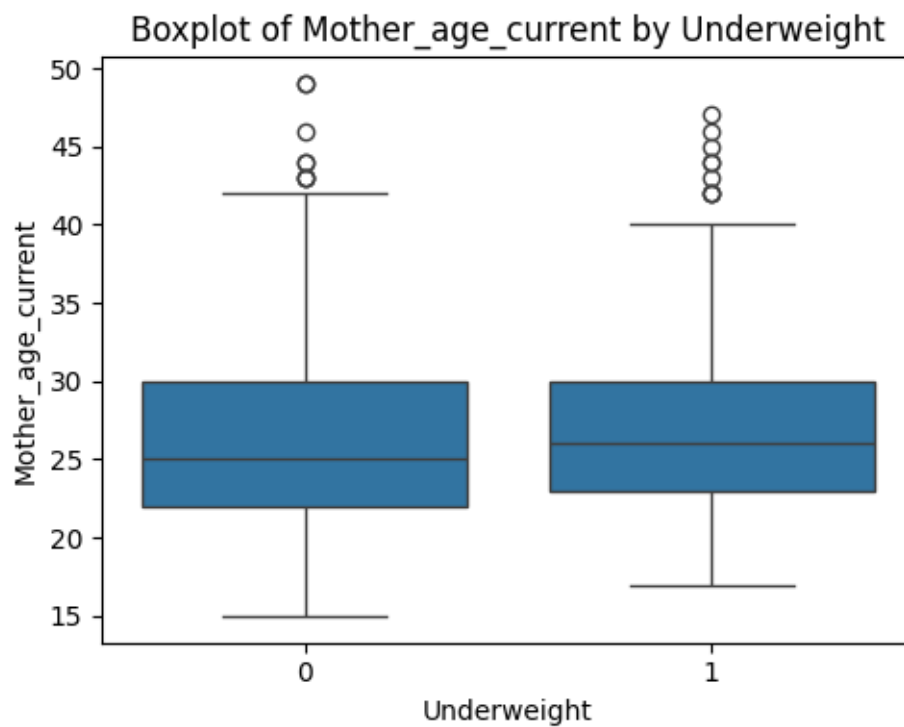
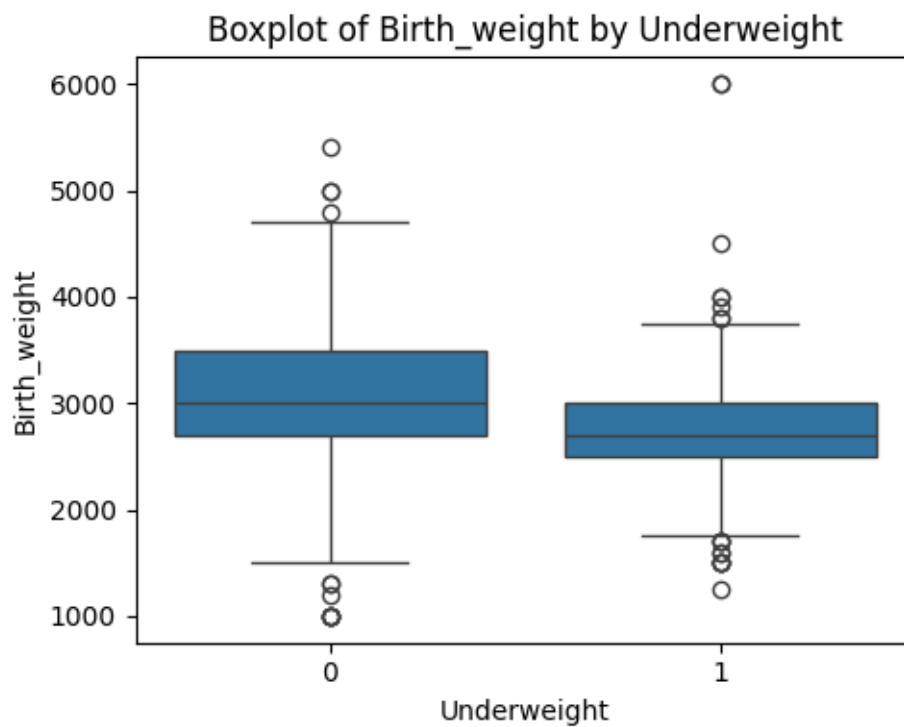
```
[13]: # Create a heatmap  
correlation_matrix = df2[numeric_cols].corr()  
  
plt.figure(figsize=(11, 7)) # Set figure size  
sns.heatmap(correlation_matrix, annot=True, cmap='coolwarm', fmt='.3f')  
plt.title("Correlation Matrix Heatmap")  
plt.show()
```

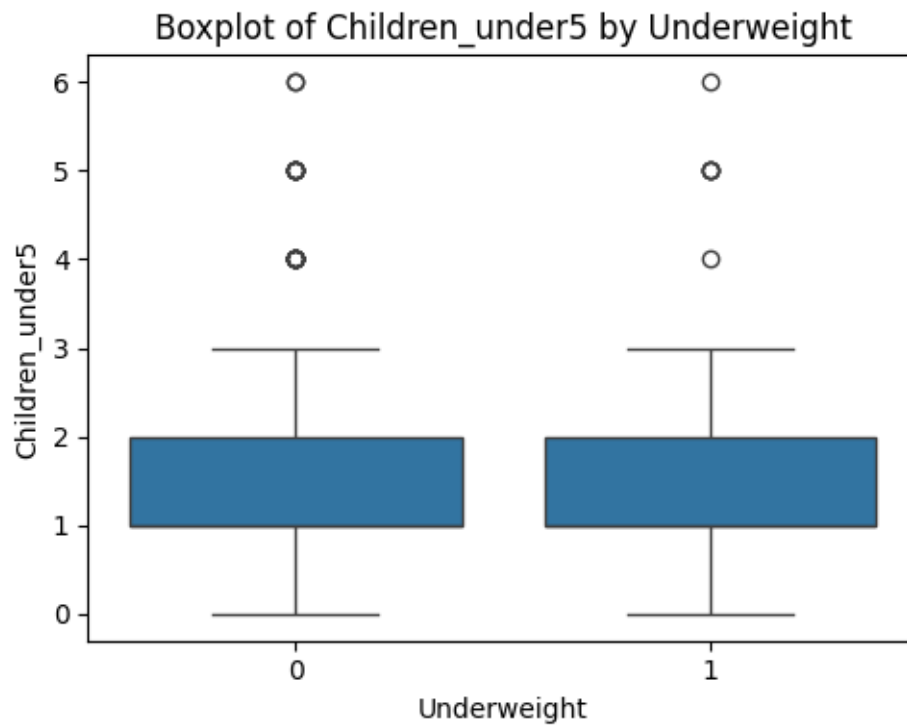
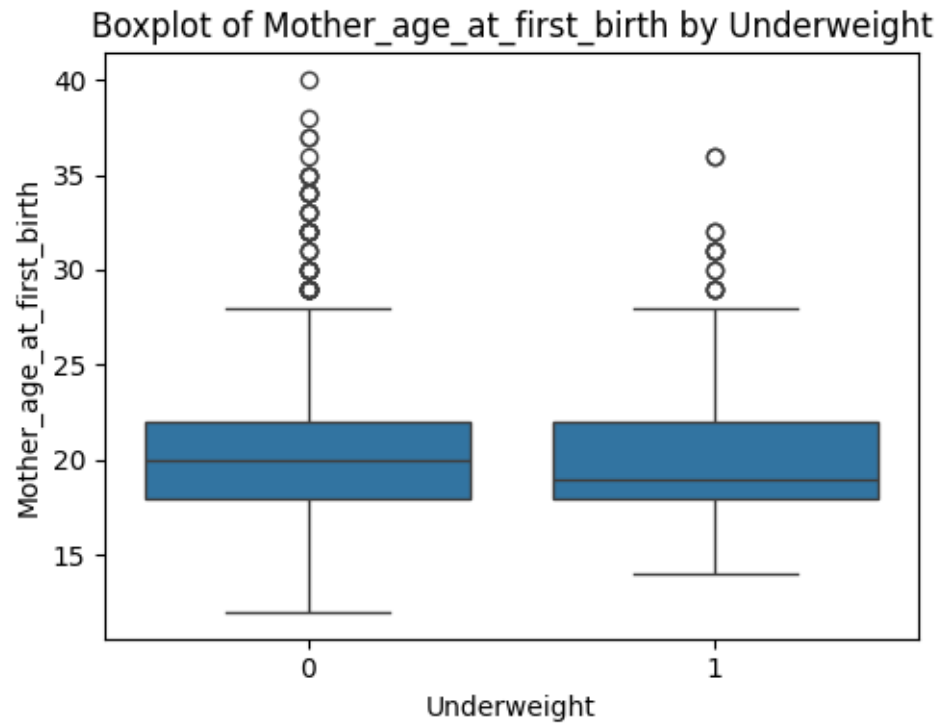


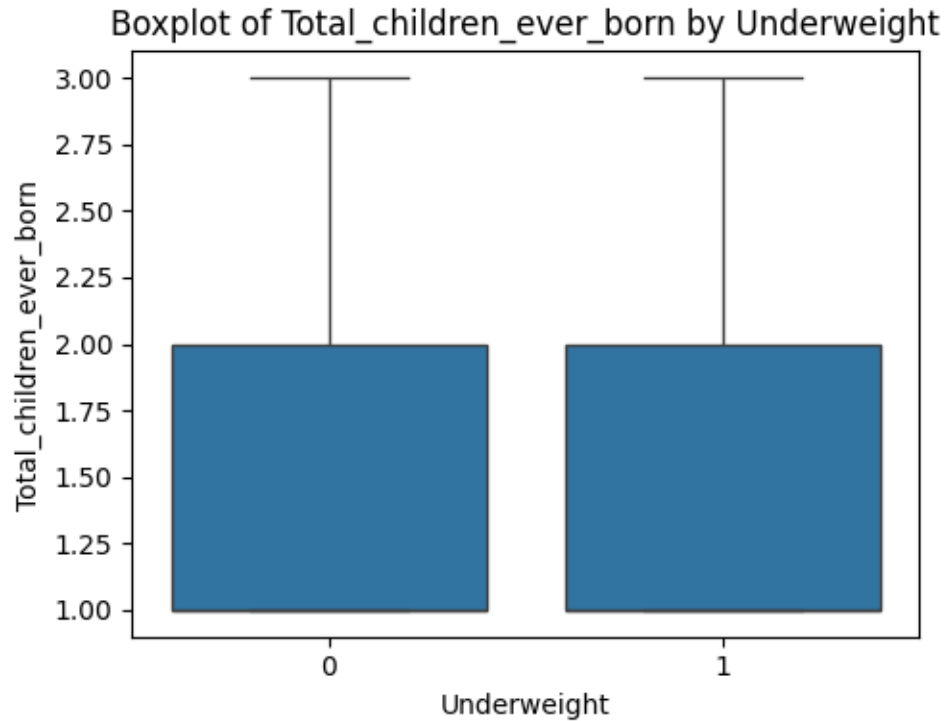
```
[14]: # Loop through each numeric column (excluding 'Underweight' itself)
for col in numeric_cols:
    if col not in ['Underweight', 'Stunting', 'Wasting']:
        plt.figure(figsize=(5, 4))
        sns.boxplot(x='Underweight', y=col, data=df2)
        plt.title(f'Boxplot of {col} by Underweight')
        plt.xlabel('Underweight')
        plt.ylabel(col)
        plt.tight_layout()
        plt.show()
```











- Age_first_sex and Mother_age_at_first_birth look multicollinear also they are not significantly associated with response variable so Age_first_sex is removed.
- Mother_age_at_first_birth and Mother_age_current are multicollinear and not closely associated with response variable so Mother_age_at_first_birth

0.2 Feature Importance Check for Categorical Features

0.2.1 χ^2 Test

```
[15]: # Listing all categorical features
cat_cols = ['Mother_education', 'Pregnancy_terminated', 'Wealth_index',
            ↪ 'Place_residence', 'Region', 'Religion', 'Ethnicity', 'Child_sex',
            ↪ 'Breastfeeding_duration', 'Father_education', 'Mother_working',
            'Health_insurance']
```

```
[16]: # Contingency Table
for col in cat_cols:
    contingency_table = pd.crosstab(df2[col], df2['Underweight'])
    print(f"Contingency Table: {col} vs Underweight\n")
    print(contingency_table)
    print("\n" + "-"*50 + "\n")
```

Contingency Table: Mother_education vs Underweight

Underweight	0	1
Mother_education		
no education	303	111
basic	642	148
secondary	804	134
higher	78	9

Contingency Table: Pregnancy_terminated vs Underweight

Underweight	0	1
Pregnancy_terminated		
no	1402	304
yes	425	98

Contingency Table: Wealth_index vs Underweight

Underweight	0	1
Wealth_index		
poorest	581	134
poorer	354	93
middle	362	82
richer	304	57
richest	226	36

Contingency Table: Place_residence vs Underweight

Underweight	0	1
Place_residence		
urban	961	193
rural	866	209

Contingency Table: Region vs Underweight

Underweight	0	1
Region		
koshi	316	48
madhesh province	297	108
bagmati province	228	33
gandaki province	159	33
lumbini province	255	71

karnali province	285	58
sudurpashchim province	287	51

Contingency Table: Religion vs Underweight

Underweight	0	1
Religion		
hindu	1576	357
buddhist	81	10
muslim	71	21
kirat	40	2
christian	59	12

Contingency Table: Ethnicity vs Underweight

Underweight	0	1
Ethnicity		
hill brahmin	133	29
hill chhetri	477	70
terai brahmin/chhetri	16	4
other terai caste	232	91
hill dalit	238	51
terai dalit	102	39
newar	38	4
hill janajati	362	55
terai janajati	158	38
muslim	71	21

Contingency Table: Child_sex vs Underweight

Underweight	0	1
Child_sex		
male	980	196
female	847	206

Contingency Table: Breastfeeding_duration vs Underweight

Underweight	0	1
Breastfeeding_duration		
ever breastfed, not currently breastfeeding	112	29

never breastfed	3	2
still breastfeeding	885	173

Contingency Table: Father_education vs Underweight

Underweight	0	1
Father_education		
no education	143	49
incomplete primary	532	128
complete primary	197	43
incomplete secondary	550	115
complete secondary	253	40
higher	132	23

Contingency Table: Mother_working vs Underweight

Underweight	0	1
Mother_working		
no	755	163
yes	1072	239

Contingency Table: Health_insurance vs Underweight

Underweight	0	1
Health_insurance		
no	1637	375
yes	190	27

```
[17]: from scipy.stats import chi2_contingency

# Dictionary to store test results
chi2_results = {}

for col in cat_cols:
    # Create contingency table
    contingency_table = pd.crosstab(df2[col], df2['Underweight'])

    # Perform Chi-squared test
```

```

chi2, p, dof, expected = chi2_contingency(contingency_table)

# Store the results
chi2_results[col] = {
    'Chi2 Statistic': chi2,
    'p-value': p,
    'Degrees of Freedom': dof
}

# Print summary
print(f"{col}: Chi2 = {chi2:.2f}, p = {p:.4f}, dof = {dof}")

```

```

Mother_education: Chi2 = 34.23, p = 0.0000, dof = 3
Pregnancy_terminated: Chi2 = 0.17, p = 0.6796, dof = 1
Wealth_index: Chi2 = 7.12, p = 0.1297, dof = 4
Place_residence: Chi2 = 2.60, p = 0.1069, dof = 1
Region: Chi2 = 36.80, p = 0.0000, dof = 6
Religion: Chi2 = 9.80, p = 0.0440, dof = 4
Ethnicity: Chi2 = 51.88, p = 0.0000, dof = 9
Child_sex: Chi2 = 2.96, p = 0.0853, dof = 1
Breastfeeding_duration: Chi2 = 3.47, p = 0.1766, dof = 2
Father_education: Chi2 = 13.22, p = 0.0214, dof = 5
Mother_working: Chi2 = 0.05, p = 0.8175, dof = 1
Health_insurance: Chi2 = 4.68, p = 0.0306, dof = 1

```

0.2.2 Cramers V Test

$$V = \sqrt{\frac{\chi^2/n}{\min(k-1, r-1)}}$$

```

[18]: from scipy.stats import chi2_contingency

cramers_v_results = {}

for col in cat_cols:
    if col != 'Underweight':
        # Contingency table
        table = pd.crosstab(df2[col], df2['Underweight'])

        # Chi-squared test
        chi2, _, _, _ = chi2_contingency(table)

        # Total sample size
        n = table.sum().sum()

        # Phi-squared
        phi2 = chi2 / n

```

```

# Degrees of freedom: min(k - 1, r - 1)
r, k = table.shape
dof = min(k - 1, r - 1)

# Cramér's V
V = np.sqrt(phi2 / dof)

cramers_v_results[col] = V
print(f"{col}: Cramér's V = {V:.4f}")

```

```

Mother_education: Cramér's V = 0.1239
Pregnancy_terminated: Cramér's V = 0.0087
Wealth_index: Cramér's V = 0.0565
Place_residence: Cramér's V = 0.0341
Region: Cramér's V = 0.1285
Religion: Cramér's V = 0.0663
Ethnicity: Cramér's V = 0.1526
Child_sex: Cramér's V = 0.0364
Breastfeeding_duration: Cramér's V = 0.0537
Father_education: Cramér's V = 0.0774
Mother_working: Cramér's V = 0.0049
Health_insurance: Cramér's V = 0.0458

```

Only Mother_education, Region, and Ethnicity show weak association with Underweight. All other variables have negligible association, based on Cramér's V.

Cramér's V Bar Plot

```

[19]: # Cramér's V results
cramers_v = {
    'Mother_education': 0.1239,
    'Pregnancy_terminated': 0.0087,
    'Wealth_index': 0.0565,
    'Place_residence': 0.0341,
    'Region': 0.1285,
    'Religion': 0.0663,
    'Ethnicity': 0.1526,
    'Child_sex': 0.0364,
    'Breastfeeding_duration': 0.0537,
    'Father_education': 0.0774,
    'Mother_working': 0.0049,
    'Health_insurance': 0.0458
}

# Sort the dictionary by values (descending)
cramers_v_sorted = dict(sorted(cramers_v.items(), key=lambda item: item[1],
    ↪reverse=True))

```



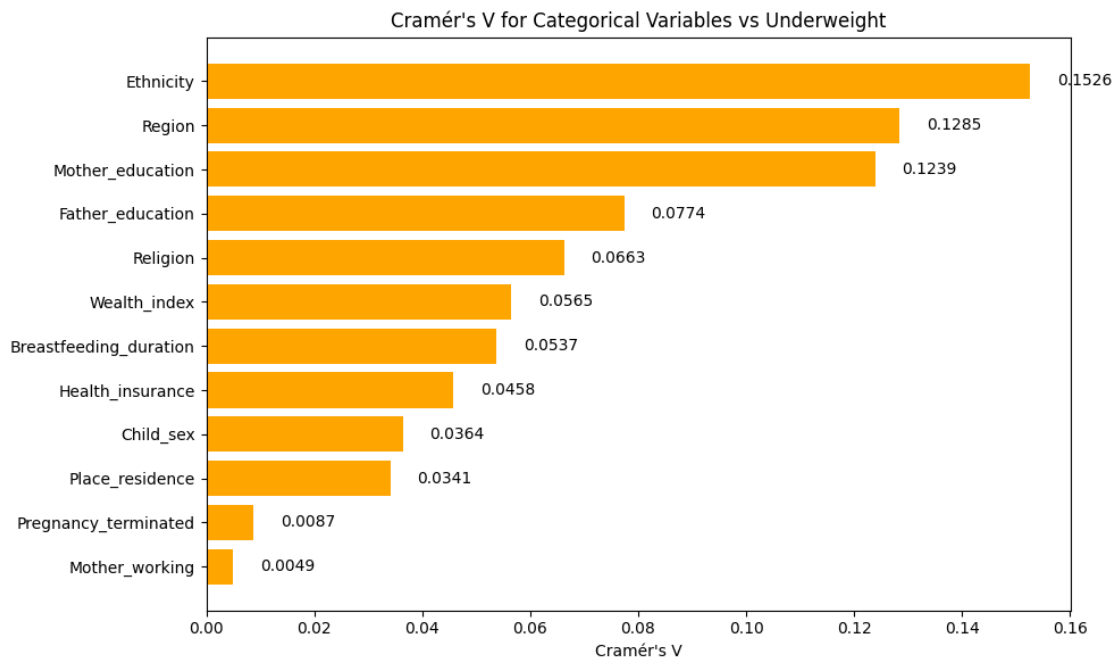
```

# Plotting
plt.figure(figsize=(10, 6))
bars = plt.barh(
    list(cramers_v_sorted.keys()),
    list(cramers_v_sorted.values()),
    color='orange'
)

# Add value labels
for bar in bars:
    width = bar.get_width()
    plt.text(width + 0.005, bar.get_y() + bar.get_height()/2,
             f'{width:.4f}', va='center')

# Aesthetics
plt.xlabel("Cramér's V")
plt.title("Cramér's V for Categorical Variables vs Underweight")
plt.gca().invert_yaxis() # Highest V at the top
plt.tight_layout()
plt.show()

```



```

[20]: df3 = df2.drop(columns=['Age_first_sex', 'Mother_age_at_first_birth',
    ↪ 'Mother_working', 'Pregnancy_terminated', 'Breastfeeding_duration',
    ↪ 'Birth_interval', 'Birth_weight'])

```

```
[21]: df3.isna().sum()
```

```
[21]: Child_age          0
      Mother_education   0
      Wealth_index       0
      Place_residence    0
      Region            0
      BMI               0
      Children_under5    0
      Total_children_ever_born 0
      Religion          0
      Ethnicity         0
      Child_sex         0
      Mother_age_current 0
      Father_education   24
      Health_insurance   0
      Underweight        0
      Stunting           0
      Wasting            0
      dtype: int64
```

```
[22]: df3 = df3.dropna()
```

```
[23]: df3.shape
```

```
[23]: (2205, 17)
```

```
[24]: ordinal_mappings = {
      'Mother_education': {
          'no education': 0,
          'basic': 1,
          'secondary': 2,
          'higher': 3
      },
      'Wealth_index': {
          'poorest': 0,
          'poorer': 1,
          'middle': 2,
          'richer': 3,
          'richest': 4
      },
      'Father_education': {
          'no education': 0,
          'incomplete primary': 1,
          'complete primary': 1,
          'incomplete secondary': 2,
          'complete secondary': 2,
      }
```

```

        'higher': 3
    }
}

for col, mapping in ordinal_mappings.items():
    df3[col] = df3[col].map(mapping)

```

```

[25]: df4 = pd.get_dummies(df3, columns=['Mother_education', 'Wealth_index',
    ↪ 'Place_residence', 'Region', 'Religion', 'Ethnicity', 'Child_sex',
    ↪ 'Father_education', 'Health_insurance'], drop_first=True)
df4 = df4.astype({col: int for col in df4.select_dtypes('bool').columns})
df4.head()

```

```

[25]:
   Child_age  BMI  Children_under5  Total_children_ever_born \
3          17  22.00              1                      1
4          40  25.10              2                      2
5          59  25.10              2                      2
11         55  21.53              1                      1
14         14  28.03              1                      1

   Mother_age_current  Underweight  Stunting  Wasting  Mother_education_1 \
3                   34            0         0         0                      1
4                   26            0         1         0                      0
5                   26            0         1         0                      0
11                  28            1         1         0                      0
14                  26            0         0         0                      1

   Mother_education_2  ...  Ethnicity_terai dalit  Ethnicity_newar \
3                   0  ...                    0                    0
4                   1  ...                    0                    0
5                   1  ...                    0                    0
11                  1  ...                    0                    0
14                  0  ...                    0                    0

   Ethnicity_hill janajati  Ethnicity_terai janajati  Ethnicity_muslim \
3                   0              0              0              0
4                   1              0              0              0
5                   1              0              0              0
11                  1              0              0              0
14                  1              0              0              0

   Child_sex_female  Father_education_1.0  Father_education_2.0 \
3                   1                    0                    1
4                   0                    1                    0
5                   1                    1                    0
11                  1                    1                    0
14                  0                    0                    0

```

	Father_education_3.0	Health_insurance_yes
3	0	0
4	0	0
5	0	0
11	0	0
14	0	0

[5 rows x 40 columns]

[26]: df4.info()

```
<class 'pandas.core.frame.DataFrame'>
Index: 2205 entries, 3 to 5369
Data columns (total 40 columns):
#   Column                                     Non-Null Count  Dtype
---  -
0   Child_age                                2205 non-null   int8
1   BMI                                      2205 non-null   float64
2   Children_under5                          2205 non-null   int8
3   Total_children_ever_born                 2205 non-null   int64
4   Mother_age_current                       2205 non-null   int8
5   Underweight                             2205 non-null   int64
6   Stunting                                2205 non-null   int64
7   Wasting                                  2205 non-null   int64
8   Mother_education_1                       2205 non-null   int64
9   Mother_education_2                       2205 non-null   int64
10  Mother_education_3                       2205 non-null   int64
11  Wealth_index_1                           2205 non-null   int64
12  Wealth_index_2                           2205 non-null   int64
13  Wealth_index_3                           2205 non-null   int64
14  Wealth_index_4                           2205 non-null   int64
15  Place_residence_rural                    2205 non-null   int64
16  Region_madhesh province                  2205 non-null   int64
17  Region_bagmati province                  2205 non-null   int64
18  Region_gandaki province                  2205 non-null   int64
19  Region_lumbini province                  2205 non-null   int64
20  Region_karnali province                  2205 non-null   int64
21  Region_sudurpashchim province            2205 non-null   int64
22  Religion_buddhist                        2205 non-null   int64
23  Religion_muslim                          2205 non-null   int64
24  Religion_kirat                           2205 non-null   int64
25  Religion_christian                       2205 non-null   int64
26  Ethnicity_hill chhetri                   2205 non-null   int64
27  Ethnicity_terai brahmin/chhetri          2205 non-null   int64
28  Ethnicity_other terai caste              2205 non-null   int64
29  Ethnicity_hill dalit                     2205 non-null   int64
```

```

30 Ethnicity_terai dalit          2205 non-null   int64
31 Ethnicity_newar               2205 non-null   int64
32 Ethnicity_hill janajati       2205 non-null   int64
33 Ethnicity_terai janajati      2205 non-null   int64
34 Ethnicity_muslim              2205 non-null   int64
35 Child_sex_female              2205 non-null   int64
36 Father_education_1.0          2205 non-null   int64
37 Father_education_2.0          2205 non-null   int64
38 Father_education_3.0          2205 non-null   int64
39 Health_insurance_yes          2205 non-null   int64
dtypes: float64(1), int64(36), int8(3)
memory usage: 661.1 KB

```

0.2.3 Variance Inflation Factor Check

```

[27]: X = add_constant(df4)

# Calculate VIF for each variable
vif = pd.DataFrame({
    "Variable": X.columns,
    "VIF": [variance_inflation_factor(X.values, i) for i in range(X.shape[1])]
})

print(vif)

```

	Variable	VIF
0	const	141.311809
1	Child_age	1.132227
2	BMI	1.279823
3	Children_under5	1.677557
4	Total_children_ever_born	1.708989
5	Mother_age_current	1.399837
6	Underweight	1.560835
7	Stunting	1.364860
8	Wasting	1.237561
9	Mother_education_1	2.531597
10	Mother_education_2	3.369423
11	Mother_education_3	1.811615
12	Wealth_index_1	1.651081
13	Wealth_index_2	1.925496
14	Wealth_index_3	1.980857
15	Wealth_index_4	2.236331
16	Place_residence_rural	1.178007
17	Region_madhesh province	2.683431
18	Region_bagmati province	1.914452
19	Region_gandaki province	1.640354
20	Region_lumbini province	1.821221
21	Region_karnali province	2.379609

```

22 Region_sudurpashchim province 2.233034
23 Religion_buddhist 1.299330
24 Religion_muslim inf
25 Religion_kirat 1.256434
26 Religion_christian 1.093600
27 Ethnicity_hill chhetri 3.689599
28 Ethnicity_terai brahmin/chhetri 1.170963
29 Ethnicity_other terai caste 3.900595
30 Ethnicity_hill dalit 2.903472
31 Ethnicity_terai dalit 2.442536
32 Ethnicity_newar 1.303344
33 Ethnicity_hill janajati 3.743457
34 Ethnicity_terai janajati 2.334628
35 Ethnicity_muslim inf
36 Child_sex_female 1.028228
37 Father_education_1.0 3.942952
38 Father_education_2.0 4.655824
39 Father_education_3.0 2.501413
40 Health_insurance_yes 1.161261

```

```

/home/codespace/.python/current/lib/python3.12/site-
packages/statsmodels/stats/outliers_influence.py:197: RuntimeWarning: divide by
zero encountered in scalar divide

```

```

vif = 1. / (1. - r_squared_i)

```

inf indicates the perfect multicollinearity between Religion_muslim and Ethnicity_muslim. There is no Ethnicity_muslim, it may be due to record error. We will remove Ethnicity_muslim from the dataframe.

```

[28]: df = df4.drop(columns=['Ethnicity_muslim'])
df.info()

```

```

<class 'pandas.core.frame.DataFrame'>

```

```

Index: 2205 entries, 3 to 5369

```

```

Data columns (total 39 columns):

```

#	Column	Non-Null Count	Dtype
0	Child_age	2205 non-null	int8
1	BMI	2205 non-null	float64
2	Children_under5	2205 non-null	int8
3	Total_children_ever_born	2205 non-null	int64
4	Mother_age_current	2205 non-null	int8
5	Underweight	2205 non-null	int64
6	Stunting	2205 non-null	int64
7	Wasting	2205 non-null	int64
8	Mother_education_1	2205 non-null	int64
9	Mother_education_2	2205 non-null	int64
10	Mother_education_3	2205 non-null	int64
11	Wealth_index_1	2205 non-null	int64

12	Wealth_index_2	2205 non-null	int64
13	Wealth_index_3	2205 non-null	int64
14	Wealth_index_4	2205 non-null	int64
15	Place_residence_rural	2205 non-null	int64
16	Region_madhesh province	2205 non-null	int64
17	Region_bagmati province	2205 non-null	int64
18	Region_gandaki province	2205 non-null	int64
19	Region_lumbini province	2205 non-null	int64
20	Region_karnali province	2205 non-null	int64
21	Region_sudurpashchim province	2205 non-null	int64
22	Religion_buddhist	2205 non-null	int64
23	Religion_muslim	2205 non-null	int64
24	Religion_kirat	2205 non-null	int64
25	Religion_christian	2205 non-null	int64
26	Ethnicity_hill chhetri	2205 non-null	int64
27	Ethnicity_terai brahmin/chhetri	2205 non-null	int64
28	Ethnicity_other terai caste	2205 non-null	int64
29	Ethnicity_hill dalit	2205 non-null	int64
30	Ethnicity_terai dalit	2205 non-null	int64
31	Ethnicity_newar	2205 non-null	int64
32	Ethnicity_hill janajati	2205 non-null	int64
33	Ethnicity_terai janajati	2205 non-null	int64
34	Child_sex_female	2205 non-null	int64
35	Father_education_1.0	2205 non-null	int64
36	Father_education_2.0	2205 non-null	int64
37	Father_education_3.0	2205 non-null	int64
38	Health_insurance_yes	2205 non-null	int64

dtypes: float64(1), int64(35), int8(3)

memory usage: 643.8 KB

```
[29]: df.to_csv("//workspaces/malnutrition/Methods/df.csv", index=False)
```